

Wandering Black Hole Feeds on Its Gravitational Wake

A 35,000-solar-mass black hole off galactic center accretes gas captured in its own gravitational wake.

arXiv:2608.10719; multiwavelength evidence for off-nuclear intermediate-mass black hole accretion.

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[arXiv:2608.10719](#) [doi:10.48550/arXiv.2608.10719](#)



Background

Intermediate-mass black holes are thought to seed supermassive black holes, but many should wander far from galactic nuclei after galaxy mergers. Without a central fuel supply, their growth mechanism has remained unclear.

Direct observational evidence that wandering black holes can accrete from gravitational wakes was missing.

What this paper is selling

The authors report a 35,000-solar-mass off-nuclear black hole accreting through a gravitational wake. Multi-epoch spectroscopy reveals upstream inflow, a dense downstream wake, and rapid broad-line variability.

Key results to remember

- Broad lines, compact continuum, and variability confirm a 35,000-solar-mass black hole.
- Three gas components trace upstream flow, downstream wake, and optically thick absorbers.
- Changing-look broad-line emission arises from absorbers within the capture radius.

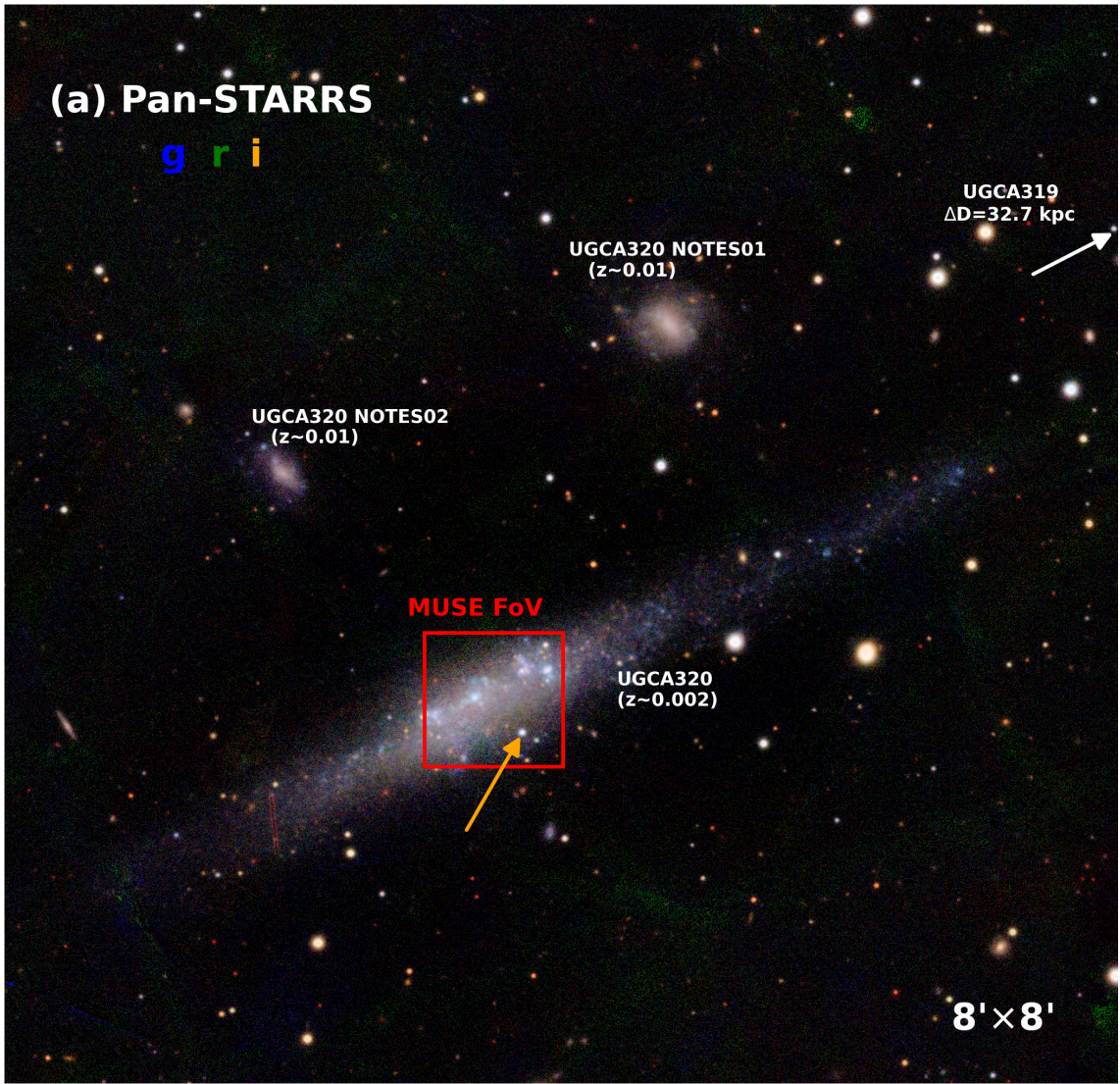


Figure 1 | Overview of UGCA 320 and UGCA320-IMBH.

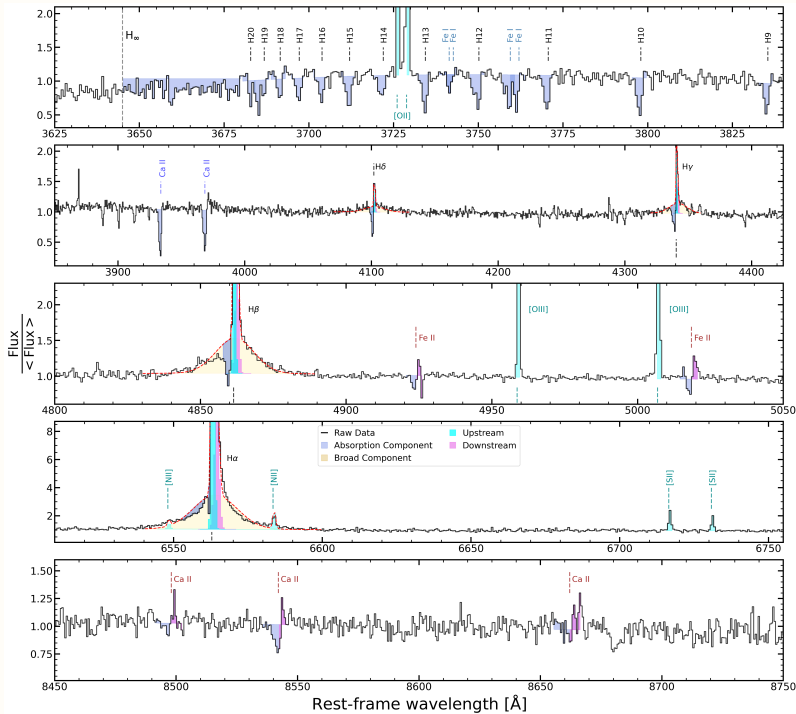


Figure 3 | X-Shooter spectrum of the UGCA320-IMBH.